

Samira Arfaee

Ph.D. Candidate in Mathematics

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Research Interests

Probability and algebraic combinatorics, with particular interest in Markov chains and mixing times on the symmetric group and its type A Iwahori–Hecke algebra, using representation-theoretic and algebraic methods.

Education

- 2021–Present **Ph.D. in Mathematics**, *Stony Brook University*, Stony Brook, NY, USA
Advisor: **Evita Nestoridi**.
- 2019–2021 **M.Sc. in Mathematics**, *École Polytechnique Fédérale de Lausanne (EPFL)*,
Lausanne, Switzerland
- 2014–2019 **B.S. in Mathematics and Applications**, *Amirkabir University of Technology (AUT)*, Tehran, Iran
Second rank.
- 2016–2019 **Minor in Electrical Engineering (Communications)**, *Amirkabir University of Technology (AUT)*, Tehran, Iran

Publications and Preprints

- 2026 *Cutoff for q -deformed classical card shuffles in the type A Iwahori–Hecke algebra*, with Bryan Wong. Manuscript in preparation.
- 2025 *Shuffling via sums of Jucys–Murphy Elements*, with Evita Nestoridi. arXiv:2504.07918.
- 2023 *A new approach to character-free proof for Frobenius theorem*, with Vahid Shahverdi. *AUT Journal of Mathematics and Computing*, Vol. 4, No. 1, pp. 99–103.

Honors and Awards

- 2026 **Chairman’s Award for Excellence in Graduate Studies**, *Stony Brook University*
For exceptional contributions to research, teaching, and service.
- 2018 **Second Prize**, *International Mathematics Competition (IMC)*, Blagoevgrad, Bulgaria

- 2018 **Third Prize**, *Al-Khorezmi International Mathematical Olympiad (AKIMO)*, Uzbekistan
- 2017 **Silver Medal**, *Iranian Mathematics Competition for University Students*
- 2016 **Bronze Medal**, *Iranian Mathematics Competition for University Students*

Professional Service

- Refereeing Referee for *The Annals of Applied Probability* and *SIAM Undergraduate Research Online (SIURO)*.

Invited Talks and Presentations

- July 2026 **4th Simons Math Summer Workshop: Algebraic Methods in Probability**, *Simons Center for Geometry and Physics*, Stony Brook, NY, USA
- Apr. 2026 **AMS Spring Central Sectional Meeting**, *North Dakota State University*, Fargo, ND, USA
- Oct. 2025 **AMS Fall Southeastern Sectional Meeting**, *Tulane University*, New Orleans, LA, USA
- Mar. 2025 **Recent Developments on Mixing Times**, *Simons Center for Geometry and Physics*, Stony Brook, NY, USA

Conferences, Workshops, and Summer Schools

- Dec. 2025 **Webs in Algebra, Geometry, Topology, and Combinatorics**, *ICERM*, Providence, RI, USA
- June-July 2025 **Graphical Models in Algebraic Combinatorics**, *Summer Graduate School, SLMath*, Berkeley, CA, USA
- Feb. 2025 **Connections Workshop: Extremal Combinatorics**, *SLMath*, Berkeley, CA, USA
- Jan. 2025 **Introductory Workshop: Probability and Statistics of Discrete Structures**, *SLMath*, Berkeley, CA, USA
- Jan. 2025 **Connections Workshop: Probability and Statistics of Discrete Structures**, *SLMath*, Berkeley, CA, USA
- July. 2024 **Virginia Integrable Probability Summer School**, *University of Virginia*, Charlottesville, VA, USA
- June. 2024 **The Many Combinatorial Legacies of Richard P. Stanley**, *Harvard University*, Cambridge, MA, USA
- Mar. 2024 **Universality & Integrability in KPZ**, *Columbia University*, New York, NY, USA
- Oct. 2023 **The Asymmetric Simple Exclusion Process**, *Simons Center for Geometry and Physics*, Stony Brook, NY, USA

Teaching Experience

Stony Brook University

- Summer 2025, **Teaching Assistant**, *Simons STEM Scholars Program*, Stony Brook University
2026 Invited to return in Summer 2026 after serving in Summer 2025.
- Fall 2025 **Lecturer**, *Trigonometry and Logarithms*, Stony Brook University
Two sections.
- Spring, Fall **Lecturer**, *Proficiency Algebra*, Stony Brook University
2024 Substitute instructor in Spring 2024.
- Fall 2023; **Lecturer / Recitation Instructor**, *Calculus I*, Stony Brook University
Summer 2024 Recitation Instructor, two sections, Fall 2023; Lecturer, Summer 2024.
- Summer 2022, **Lecturer / Recitation Instructor**, *Calculus A*, Stony Brook University
Spring 2023, Lecturer, Summer 2022 and Summer 2023; Recitation Instructor, two sections, Spring
Summer 2023 2023.
- Fall 2022 **Recitation Instructor**, *Linear Algebra*, Stony Brook University
Two sections.
- Summer 2022 **Lecturer**, *Calculus III and Applications*, Stony Brook University
- Fall 2021 **Recitation Instructor**, *Calculus B*, Stony Brook University
Two sections.

Earlier Teaching

- Fall 2020 **Teaching Assistant**, *Discrete Mathematics*, EPFL, Lausanne, Switzerland
- Fall 2017 **Teaching Assistant**, *Foundations of Mathematical Analysis*, Amirkabir University of Technology, Tehran, Iran
- Spring, Fall **Teaching Assistant**, *Foundations of Matrix and Linear Algebra*, Amirkabir
2017 University of Technology, Tehran, Iran
- Spring 2017 **Teaching Assistant**, *Foundations of Algebra*, Amirkabir University of Technology, Tehran, Iran

Mentoring

- Fall 2025 **Mentor**, *Directed Reading Program*, Stony Brook University
Mentored undergraduate students in probability, group theory, and Markov chains; guided independent reading, proof-writing, and presentations.

Languages

Persian (Farsi): Native **English:** Fluent

Interests

Rock climbing and solving puzzles, especially jigsaw puzzles.